Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions

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Dataset Overview

3,900

Purchase Records

Transactional data analyzed

18

Data Fields

Customer and purchase details

37

Missing Values

Review ratings imputed

Key features include customer demographics, purchase details, and shopping behavior metrics.

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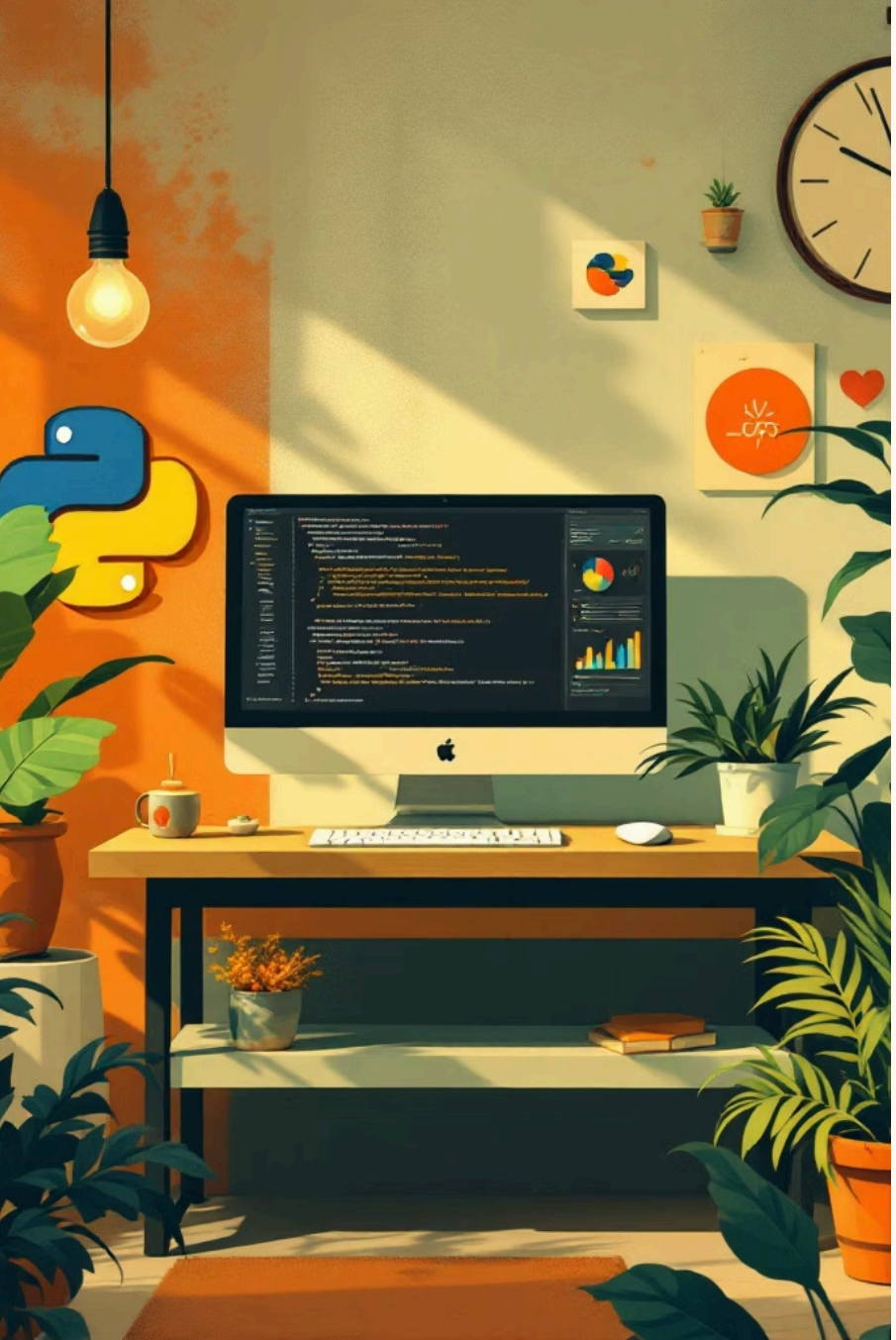

Python Data Preparation

Data Cleaning

- Imported dataset using pandas
- Checked structure with info() and describe()
- Imputed missing review ratings using median by category

Feature Engineering

- Created age_group bins
- Calculated purchase_frequency_days
- Dropped redundant promo_code_used column

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SQL Analysis: Revenue Insights

Gender Revenue

Male customers generated $157,890 vs. Female $75,191

Subscription Impact

Non-subscribers spent more on average ($59.87) vs. subscribers ($59.49)

Shipping Preference

Express shipping customers spent $60.48 vs. Standard $58.46

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Product Performance

Top Rated

Gloves: 3.86 avg rating

Most Discounted

Hats: 50% discount rate

Best Seller

Blouse: 171 orders

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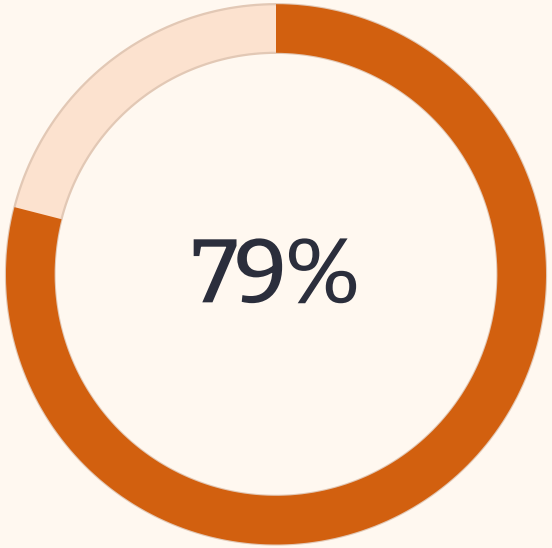
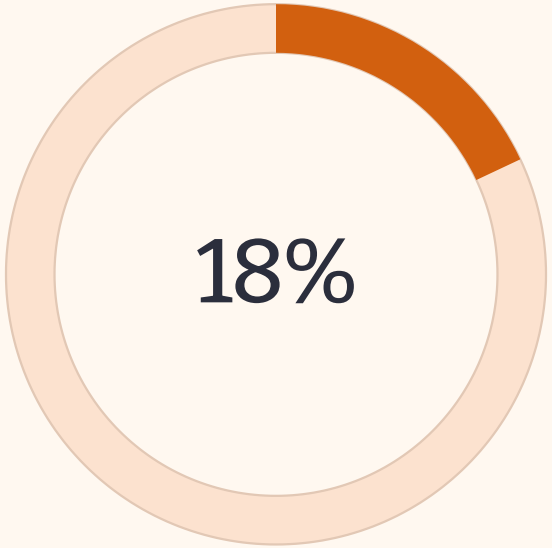
Customer Segmentation



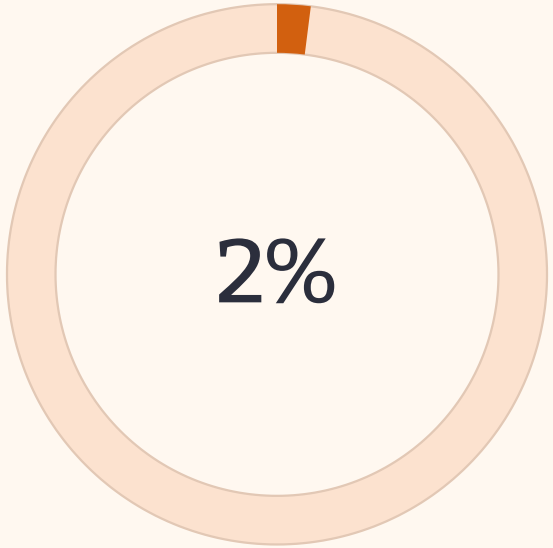
Loyal Customers

3,116 customers with consistent purchase history



Returning

701 customers making second purchases



New

83 first-time customers

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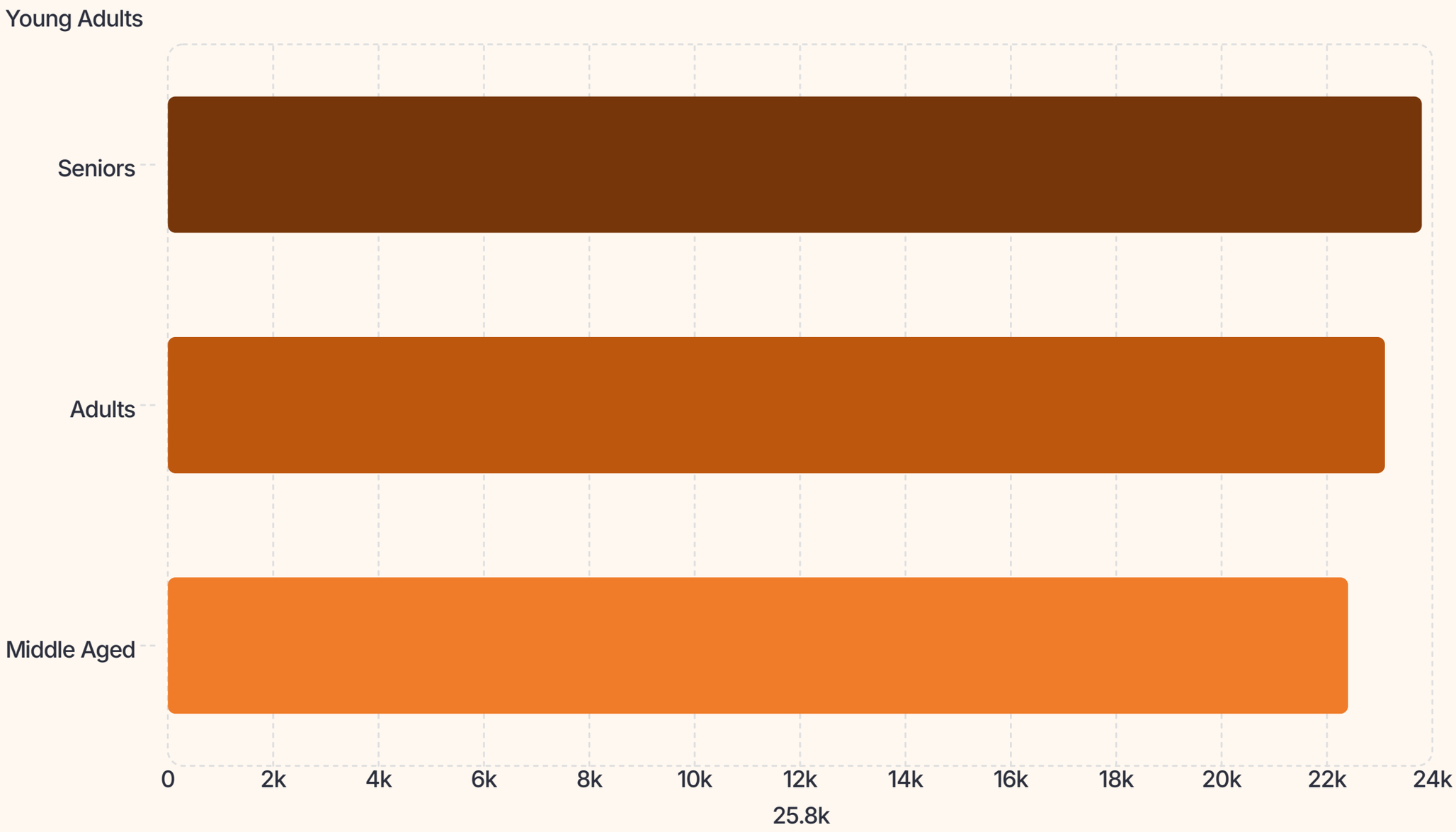
Revenue by Age Group



Young adults generated the highest revenue at $62,143, followed by middle-aged customers at $59,197.

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Power BI Dashboard

01

KPIs Display

3.9K customers, $59.76 avg purchase, 3.75 avg rating

02

Interactive Filters

Subscription, gender, category, shipping type

03

Visual Charts

Revenue and sales by category and age group

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Key Findings

Male Dominance

Male customers generated 2.1x more revenue than female customers ($157,890 vs $75,191)

Discount Strategy

839 customers used discounts while spending above average purchase amount

Repeat Purchase Link

958 repeat buyers (over 5 purchases) vs 2,518 single purchase customers

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Business Recommendations

Boost Subscriptions

Promote exclusive benefits for subscribers

Loyalty Programs

Reward repeat buyers to move them into loyal segment

Review Discount Policy

Balance sales boosts with margin control

Targeted Marketing

Focus on high-revenue age groups and express-shipping users

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